A patient found to have an abnormally low number of helper T cells in her blood contracts a bacterial infection. During the infection, which of the following is most likely NOT affected by the patient's low count of helper T cells?

- A. Expression of major histocompatibility complex proteins that present bacterial antigens on the membrane of infected host cells [47%]
- B. Proliferation of B lymphocytes in response to the bacterial infection [19%]
- C. Levels of circulating antibodies against the invading bacterium [17%]
- D. Activation of apoptosis-inducing cytotoxic T cells [15%]

Correct
47% answered correctly

Explanation:

Immune cell assisted by helper T cell	Helper T cell function
B lymphocyte	• Promotes B lymphocyte activation and proliferation • Induces differentiation of B lymphocytes in antibody-secreting plasma cells and memory cells
Cytotoxic T cell	• Promotes cytotoxic T cell activation and proliferation
Macrophage	• Enhances activity of macrophages
Other immune cells	• Attracts other immune cells to site of infection

The immune response comprises both innate and adaptive immunity. Cells of the innate immune system can rapidly and nonspecifically recognize and destroy foreign antigens. In contrast, the adaptive immune system, subdivided into **cell-mediated** and **humoral immunity**, comprises cells that recognize *specific* antigens and mount more specialized immune responses against invading pathogens.

In cell-mediated immunity, T lymphocytes (cells) recognize and mount immune responses against *foreign* antigens displayed by other cells on specialized surface proteins called **major histocompatibility complex (MHC) proteins**. MHC proteins on any given cell display fragments of any proteins present within that cell. Accordingly, cells containing foreign pathogens, such as certain immune cells or infected cells, will generally display foreign protein fragments (antigens) on their MHC proteins.

Helper T cells are T cells that recognize foreign antigens displayed by MHC proteins of other immune cells, such as:

- B lymphocytes, cells that bind and engulf a foreign antigen. The antigen is broken down into fragments within the B lymphocyte, and these fragments are transported to be displayed on MHC proteins present on the cell membrane. A *helper T cell* binds the foreign antigen presented by the B lymphocyte and releases signaling molecules (cytokines) that induce division of the B lymphocyte into many identical cells, some of which differentiate to **secrete antibodies against the invading pathogen**.

- **Macrophages and dendritic cells**, which engulf foreign antigens (via phagocytosis) and present these antigen fragments on the MHC proteins found on their cell membrane. Upon binding these antigens, *helper T cells* release cytokines that enhance other immune responses, including the phagocytotic activity of macrophages and the activation and proliferation of cytotoxic T cells, immune cells that release toxins to induce apoptosis in infected cells.

In this scenario, a patient with an abnormally **low number of helper T cells** contracts an infection. Consequently, a low count of helper T cells *would affect* the activation of cytotoxic T cells and B lymphocytes (which produce antibodies against the bacterium) **(Choices B, C, and D)**. However, because expression of MHC proteins is solely dependent on a cell's transcriptional and translational machinery, cells would still be able to display bacterial antigens on their MHC proteins regardless of helper T cell count.

Educational objective:
Helper T cells bind foreign antigens presented by other immune cells and release signaling molecules that enhance immune responses, such as cytotoxic T cell activation and antibody production by B lymphocytes.

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